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**CHEMISTRY**

**0620/61**

Paper 6 Alternative to Practical

**May/June 2019**

MARK SCHEME

Maximum Mark: 40

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

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This document consists of **7** printed pages.

**PUBLISHED****Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

**GENERIC MARKING PRINCIPLE 1:**

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

**GENERIC MARKING PRINCIPLE 2:**

Marks awarded are always **whole marks** (not half marks, or other fractions).

**GENERIC MARKING PRINCIPLE 3:**

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

**GENERIC MARKING PRINCIPLE 4:**

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

**GENERIC MARKING PRINCIPLE 6:**

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

| Question | Answer                                                            | Marks |
|----------|-------------------------------------------------------------------|-------|
| 1(a)     | (conical) flask (1)<br>gas jar (1)                                | 2     |
| 1(b)     | no bung in second flask (1)<br>gas jar should not be inverted (1) | 2     |
| 1(c)     | to dry the gas / remove water                                     | 1     |
| 1(d)     | litmus (1)<br>turns white/bleaches (1)                            | 2     |
| 1(e)     | chlorine / gas is poisonous / toxic                               | 1     |

| Question | Answer                                                                                                                                                                                                | Marks    |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(a)     | table of results<br><br>volumes of aqueous sodium carbonate boxes completed correctly 1, 2, 4, 5, 6, 7, 8 (1)<br><br>heights of solid boxes completed 12, 24, 48, 60, 73, 73, 73 (1)<br><br>in mm (1) | <b>3</b> |
| 2(b)     | x-axis labelled as “volume of aqueous sodium carbonate / cm <sup>3</sup> (1)<br><br>all 7 points plotted correctly (2)<br><br>two intersecting straight line graphs drawn with a ruler (1)            | <b>4</b> |
| 2(c)     | working shown on graph in correct place (1)<br><br>value from graph (1)                                                                                                                               | <b>2</b> |
| 2(d)     | height increases / proportional to volume / more solid (1)<br><br>level off / becomes constant (1)                                                                                                    | <b>2</b> |
| 2(e)     | same heights / at 73 mm (1)<br><br>all barium nitrate reacted (1)                                                                                                                                     | <b>2</b> |
| 2(f)     | use burette / pipette to measure out aqueous barium nitrate / instead of measuring cylinder                                                                                                           | <b>1</b> |
| 2(g)     | filter (1)<br><br>dry (1)<br><br>weigh solid (1)                                                                                                                                                      | <b>3</b> |
| 2(h)     | repeat <b>and</b> compare                                                                                                                                                                             | <b>1</b> |

| Question | Answer                                                                       | Marks    |
|----------|------------------------------------------------------------------------------|----------|
|          | <b>Tests on solution F</b>                                                   |          |
| 3(a)     | 0–3                                                                          | <b>1</b> |
| 3(b)     | bubbles / fizz / effervescence (1)<br>lighted splint / flame (1)<br>pops (1) | <b>3</b> |
| 3(c)     | white precipitate                                                            | <b>1</b> |
| 3(d)     | no reaction/change                                                           | <b>1</b> |
|          | <b>Tests on solid G</b>                                                      |          |
| 3(e)     | calcium (1)<br>carbonate (1)                                                 | <b>2</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                         | Marks        |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 4        | any <b>six</b> from:<br>weigh nails / use set number of nails<br>place nails in a suitable container<br>add water to nails<br>leave in water for a suitable time ( $\geq 1$ week)<br>observe nails / compare / reweigh nails<br>repeat with nail with other coverings<br>nail with least rust / mass increase has best coating | <b>Max 6</b> |